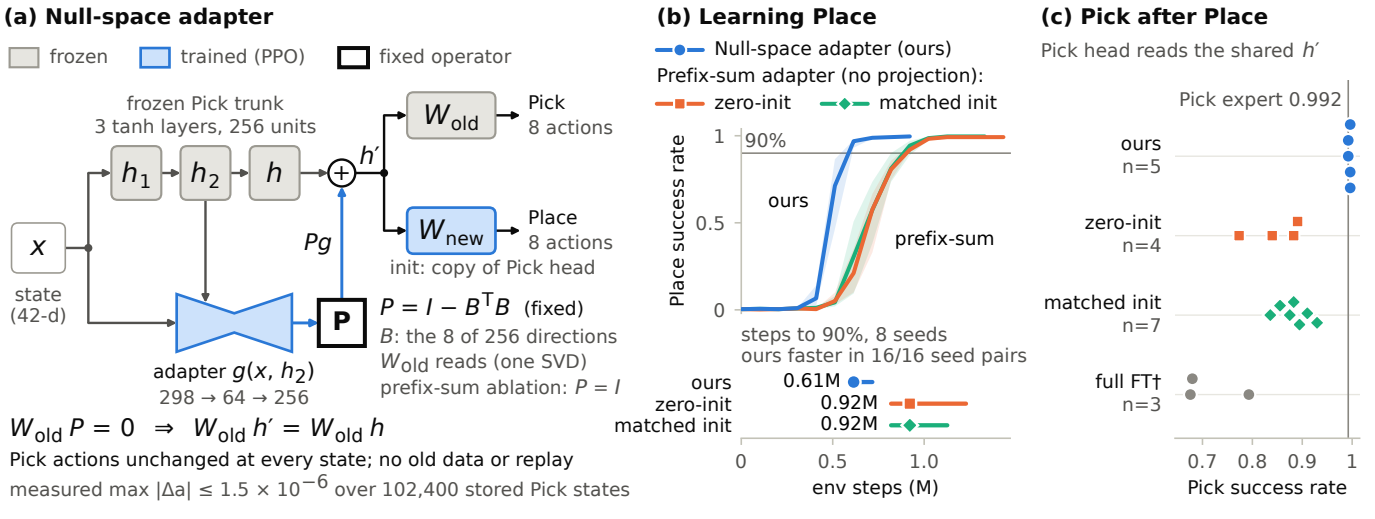
(a) Null-space adapter
frozen
trained (PPO)
fixed operator
frozen Pick trunk
3 tanh layers, 256 units
h_1
h_2
h
h'
W_{old}
Pick
8 actions
W_{new}
Place
8 actions
init: copy of Pick head
x
state
(42-d)
Pg
P
$P = I - B^{T}B$ (fixed)
B: the 8 of 256 directions
W_{old} reads (one SVD)
prefix-sum ablation: $P = I$
adapter $g(x, h_2)$
298 → 64 → 256
$W_{old}\,P = 0 \;\Rightarrow\; W_{old}\,h' = W_{old}\,h$
Pick actions unchanged at every state; no old data or replay
measured max $|\Delta a| \le 1.5 \times 10^{-6}$ over 102,400 stored Pick states
(b) Learning Place
Null-space adapter (ours)
Prefix-sum adapter (no projection):
zero-init
matched init
Place success rate
1
0.5
0
90%
ours
prefix-sum
steps to 90%, 8 seeds
ours faster in 16/16 seed pairs
ours 0.61M
zero-init 0.92M
matched init 0.92M
0
0.5
1.0
env steps (M)
(c) Pick after Place
Pick head reads the shared h'
Pick expert 0.992
ours
n=5
zero-init
n=4
matched init
n=7
full FT†
n=3
0.7
0.8
0.9
1
Pick success rate